TABLE I: Comparison of Accuracy and Communication Cost

| | Datasets | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MNIST | | | FMNIST | | | CIFAR-10 | | | SVHN | | |
| Methods | AC (%) | DV (GB) | UV (GB) | AC (%) | DV (GB) | UV (GB) | AC (%) | DV (GB) | UV (GB) | AC (%) | DV (GB) | UV (GB) |
| FedAvg+Ckks | 90.00 | **0.60** | 0.62 | 80.00 | 3.45 | 1.39 | **96.00** | 0.85 | 0.90 | 76.00 | 1.37 | 1.06 |
| FedPHE | 76.00 | 1.30 | 1.33 | 88.00 | 1.38 | 1.22 | 94.00 | **0.53** | 1.63 | 92.00 | 1.86 | **0.40** |
| ESAFL | 98.00 | 1.85 | **0.28** | 77.00 | 3.89 | 1.25 | 94.00 | 3.42 | 1.12 | **98.00** | 2.01 | 1.15 |
| MASER | 95.00 | 2.97 | 1.74 | 89.00 | 3.32 | **0.19** | 80.00 | 1.66 | **0.25** | 81.00 | 0.90 | 0.63 |
| ECPFL-UN | 90.00 | 1.96 | 0.80 | 80.00 | 1.57 | 1.88 | 91.00 | 2.94 | 0.43 | 92.00 | 1.15 | 0.82 |
| ECPFL-TO-0.3 | **99.00** | 3.06 | 1.16 | 91.00 | 3.87 | 1.57 | 80.00 | 0.63 | 0.70 | 81.00 | 1.34 | 1.89 |
| ECPFL-TO-0.5 | 96.00 | 1.76 | 1.35 | 84.00 | 4.16 | 0.97 | 81.00 | 1.49 | 1.17 | 81.00 | 2.84 | 1.81 |
| ECPFL-TH-0.3 | 85.00 | 1.38 | 2.00 | 87.00 | 0.86 | **0.19** | 78.00 | 3.01 | 1.60 | 85.00 | **0.75** | 0.83 |
| ECPFL-TH-0.5 | **99.00** | 2.62 | 1.95 | **96.00** | **0.55** | 1.47 | 91.00 | 2.65 | 0.61 | 90.00 | 0.95 | 0.93 |

# Table Preview – Communication Cost Comparison